The bag that copies

Mathematics before the session

Start with one red and one blue counter. At each step choose uniformly among the counters now present, return the chosen counter, and add one new counter of its color. After n draws the bag has n+2 counters. If r of the draws were red, its composition is (r+1) red and (n-r+1) blue. Every r from 0 through n is possible, and neither color can disappear.

This is the one-copy Pólya urn. At every fixed length n, the number of red draws is **uniform on 0,1,...,n**: each total has probability 1/(n+1). This does not say that color words are equally likely. A particular length-n word with r reds has probability r!(n-r)!/(n+1)!, independent of their order. There are n!/[r!(n-r)!] such words, giving the stated uniform total.

Counter identities make a finite proof visible. At successive draws there are 2, 3, ..., n+1 identities available. A history specifies the particular counter drawn each time, with new copies labeled by their birth step. All (n+1)! such histories are equally likely, although the available identities depend on the earlier draws. For n=2 there are six histories, with two histories each for zero, one, or two red draws. For n=3 there are 24 histories, six per total.

The important distinctions

A color word such as RR merges two two-draw identity histories; RB has only one. Thus the four words RR, RB, BR, BB have probabilities 1/3, 1/6, 1/6, 1/3. If instead the original RB bag is restored after every draw and no copy is added, those four color words each have probability 1/4, and the red-draw totals have probabilities 1/4, 1/2, 1/4.

The copying draws are dependent: after red, the next red chance is 2/3; after blue, it is 1/3. Yet reordering a word without changing its color totals does not change its probability. This exchangeability follows from multiplying the same red factors 1 through r and blue factors 1 through n-r over the same denominators 2 through n+1. Do not confuse it with independence.

All probability claims assume uniform selection of current identities at each step and the stated initial bag and addition rule. Deliberately choosing a legal word proves possibility, not its probability. Trial frequencies can inspire the equal-total conjecture but do not prove it. No claim about eventual loss of a color follows; both original counters remain at every finite time.

What each group is discovering

K-1: changes of composition, multiple stories with the same finish, and working backward one draw. **Grades 2-3:** color stories versus equally likely marked histories, and the contrast with return-only sampling. **Grades 4-5:** exact total distributions, equal-count reordering, and a general conjecture supported by a finite counting proof.

Running the session

Materials and readiness

For eleven children, prepare six opaque bags/cups and, per pair, six red plus six blue counters of equal size and feel. Only one of each begins inside. Keep the supply outside. Use removable identity labels: original counters R1 and B1; the first new copy is R2 or B2; the next is R3 or B3. The digit records when the copy was added, not the number of that color. A marker and small paper slips suffice.

Print K-1 pages 1-4, grades 2-3 pages 1-5 and grades 4-5 pages 1-4, and supply pencils plus a small tray for each ending bag. K-1 requires counting to six, color matching, and an adult reading the questions. Middle work requires tracking two draws and distinguishing identities. Upper work uses branching counts and small products; factorial notation is optional. Have an adult enact the first copied draw and supervise resetting to exactly R1/B1 before every fresh story. Prepared, unpiloted material.

Short shared launch

Let the group inspect RB. Deliberately take R out, visibly put that same R back, and add a second red: RB → draw R → RRB. Write the story in a strip as you go, one color per cell. Do a second draw so children see that both red identities can be selected. Ask, “What bags could be here after two draws?” Announce when children are choosing a possible story and when a cup is being used for a random draw.

A flexible hour

0-5 minutes: running around. **5-10:** handle counters and demonstrate returning plus copying. **10-22:** choose short stories and compare final bags; the random games on page 6 add about 15 minutes. **22-33:** K-1 extends or reverses stories; older groups construct the six marked two-draw histories. **33-35:** movement break: one child is the original counter, another joins as its copy. **35-47:** compare return-only sampling or investigate three/four draws. **47-53:** share an exact count or an invariant with the group. **53-56:** restore the starting kits.

A complete six-history argument is a strong endpoint. Do not insist that K-1 compute probabilities or that the oldest children enumerate 120 rows. Let the upper group use the 24-history result to predict the next stage. An unexplained five-draw guess should be recorded as a conjecture until justified.

Optional hints

For final bags: “How many reds were added?” For the marked tree: “After the first draw, point to each counter that could be chosen next.” For reordered words: “Which multipliers change, and which only move?” For backward work: “Which one counter could have been the last new copy?”

Source and adaptation

Janko Gravner, UC Davis MAT 235B, *Martingale convergence theorems*, February 2, 2024, slides 12-14, “Pólya’s urn”; university lecture PDF. The source gives the urn and uniform count law. Marked-counter histories, short-story tasks, and the exact finite keys here are independently derived classroom adaptations; no martingale theorem is a child prerequisite.

K to 1 answer key

Problem 1

After two draws the possible bags are **3R/1B** from RR, **2R/2B** from RB or BR, and **1R/3B** from BB. Only the middle bag has two different two-draw *color* stories. A child who counts marked counters instead finds two stories for every bag (R1 R1 or R1 R2; R1 B1 or B1 R1; B1 B1 or B1 B2): accept that, then ask which bags have two different color orders. This is a possibility question; counting these four words as equally likely would be wrong for random copying.

Problem 2

For the printed choices: 4R/1B is possible via RRR; 2R/3B via RBB, BRB, or BBR; 5R/0B is impossible; 1R/4B via BBB. The full collection of possible three-draw bags also includes 3R/2B, via RRB, RBR, or BRR: ask for it as a follow-up. The all-red picture is impossible because the original blue is never removed.

Problem 3

The displayed 3R/3B finish requires exactly two red draws and two blue draws. All six color stories are **RRBB, RBRB, RBBR, BRRB, BRBR, BBRR**. The page has eight story rows, two more than needed. Completeness can be seen by choosing the two red positions among four: (1,2), (1,3), (1,4), (2,3), (2,4), (3,4).

The four-draw bags with more blue than red are exactly **1R/5B**, made by BBBB, and **2R/4B**, made by RBBB, BRBB, BBRB, or BBBR. The bag has six counters, with at least one of each color; the other mixtures are tied or red-majority. The two bag boxes, each with a story row, hold these. Children need a story for each blue-majority bag, not necessarily all four stories for the second bag.

Problem 4

Four-draw finish	Every possible bag before draw 4	Last color
5R / 1B	4R / 1B	R
4R / 2B	3R / 2B 4R / 1B	R B
3R / 3B	2R / 3B 3R / 2B	R B

Each last draw adds exactly one counter of its own color, so remove one candidate last copy to find a predecessor. A predecessor must still contain at least one red and one blue. For example, removing the only blue from 5R/1B would make 5R/0B, which cannot have arisen from the required start.

Copying cannot lose all blue counters, or all red counters. Every drawn counter is returned, and each move only adds a counter. In particular, the original blue remains. This is a deterministic invariant, not an observation about which color seems to be winning.

Grades 2 to 3 answer key

Problem 1

RR → 3R/1B; RB → 2R/2B; BR → 2R/2B; BB → 1R/3B. The balanced finish has two color stories; each extreme has one. Color-story counts alone do not measure probability here.

Problem 2

All six marked histories are listed below. Each first choice has chance 1/2; each second choice, conditional on the first, has chance 1/3. Thus every row has chance 1/6.

First identity	Second identity	Final colors	Red draws
R1	R1	3R / 1B	2
R1	R2	3R / 1B	2
R1	B1	2R / 2B	1
B1	R1	2R / 2B	1
B1	B1	1R / 3B	0
B1	B2	1R / 3B	0

All three compositions have two histories, so none has the most: each has probability 1/3. Notice that RR and BB each merge two marked histories, while RB and BR each merge one.

Problem 3

Copying: zero, one, two red draws are tied at 2/6 = 1/3 each. **Return only:** RR, RB, BR, BB are now four equally likely histories, so the totals have weights 1, 2, 1 and probabilities 1/4, 1/2, 1/4. One red draw is most likely in the unchanged bag. Copying increases the chance of repeating the first color.

Problem 4

The stories are RRB, RBR, BRR. Each has chance 1/12: RRB gives (1/2)(2/3)(1/4); RBR gives (1/2)(1/3)(2/4); BRR gives (1/2)(1/3)(2/4). Each represents two of the 24 marked histories.

Problem 5

5R/1B: only RRRR, because all four added counters must be red. **4R/2B:** four stories, RRRB, RRBR, RBRR, BRRR. **3R/3B:** six stories, RRBB, RBRB, RBBR, BRRB, BRBR, BBRR, one for each choice of the two red draws.

Problem 6

Neither: each has chance 1/5. RRRR has 1 × 2 × 3 × 4 = 24 of the 120 marked histories, since its red draws have 1, 2, 3 and 4 red counters to choose from. Each of the six 3R/3B color stories has 1 × 2 × 1 × 2 = 4, so that bag also has 24. An adult can carry the multiplication.

Grades 4 to 5 answer key

Problem 1

The six marked histories are (R1,R1), (R1,R2), (R1,B1), (B1,R1), (B1,B1), (B1,B2). Each has probability 1/6. The first two have two red draws, the middle two have one, and the last two have zero. All totals are tied, two histories each.

Problem 2

Without copies, the four histories are RR, RB, BR, BB, each with probability 1/4. The probabilities for zero, one, two reds become 1/4, 1/2, 1/4, compared with 1/3, 1/3, 1/3 under copying. In copying, the first color gains an extra identity for the second draw, giving RR and BB two marked histories each.

Problem 3

RRB, RBR, BRR each has probability 1/12. Their products are (1/2)(2/3)(1/4), (1/2)(1/3)(2/4), and (1/2)(1/3)(2/4). Reordering does not change the product of the red factors and blue factors. The draws remain dependent despite this equality.

Problem 4

There are six marked histories for each total 0, 1, 2, 3. BBB and RRR each have $1 \times 2 \times 3 = 6$ identity histories. For one red, the three words RBB, BRB, BBR have two histories each, hence six. The three words with two reds also have two each. These four groups cover all eight possible color words and all 24 marked histories.

Problem 5

Each of the five totals has 24 of the 120 marked histories, hence chance 1/5. Extend the six histories in each three-draw group by the five available fourth-draw identities. The contributions to totals 0 through 4 are:

Final red total	From same total by blue	From one less by red	Total
0	6 × 4	0	24
1	6 × 3	6 × 1	24
2	6 × 2	6 × 2	24
3	6 × 1	6 × 3	24
4	0	6 × 4	24

Problem 6

Predict 1/6 for each red total after five draws, and $1/(n+1)$ after n draws. Proof: a word with r reds has probability $r!(n-r)!/(n+1)!$, since its red numerators are 1,...,r and its blue numerators 1,...,n-r. There are $n!/[r!(n-r)!]$ such words. Multiplication gives $1/(n+1)$. A child’s observed pattern remains a conjecture until justified; the general proof is an optional endpoint invited by the problem.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each table and reset every fresh story to the same two original counters. First routes: K-1 P1-2; grades 2-3 P1-2; grades 4-5 P1, then P2-3. Before the first chance question (grades 2-3 P2, grades 4-5 P1), pairs draw without looking for about ten two-draw copying games, then about ten return-only games, and pool the final bags on a class chart: a conjecture for the counting to test. The games take about 15 minutes. A chart of 20 pooled games of each kind shows fewer two-red, two-blue bags for copying than for returning only about four times in five, so a chart that does not is ordinary chance; the exact count decides. On return, K-1 can find four-draw stories or work backward one step (P3-4). Grades 2-3 can compare return-only sampling and reordered words (P3-4), then larger final bags (P5-6). Grades 4-5 can make the 24-history argument (P4), the four-draw count (P5), or pursue the arbitrary-length conjecture (P6). Keep exact history counts distinct from trial frequencies.